WASP-3b

R-1235

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-3_160617` · created 2026-07-31T10:21:47+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457556.8039 +/- 0.0023 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.004 +/- 0.04
Transit depth (Rp/Rs)^2	0.0016 +/- 0.032 [%]
Orbital Inclination (inc)	81.63 +/- 2.89
Airmass coefficient 1 (a1)	1.72 +/- 1.04
Airmass coefficient 2 (a2)	-0.41 +/- 0.96
Scatter in the residuals of the lightcurve fit is	57.62 %
Best Comparison Star	#1 - [554, 448]
Optimal Aperture	4.94
Optimal Annulus	5.37
Transit Duration (day)	0.076 +/- 0.035

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.004 ± 0.04 vs 0.10538 (deviation 2.43σ)
Duration fit vs published	0.076 d vs 0.1167 d (deviation 1.12σ)
Mid-time O–C	2.0 min
Depth SNR	0.1
Residual scatter	57.62 %

Light curve

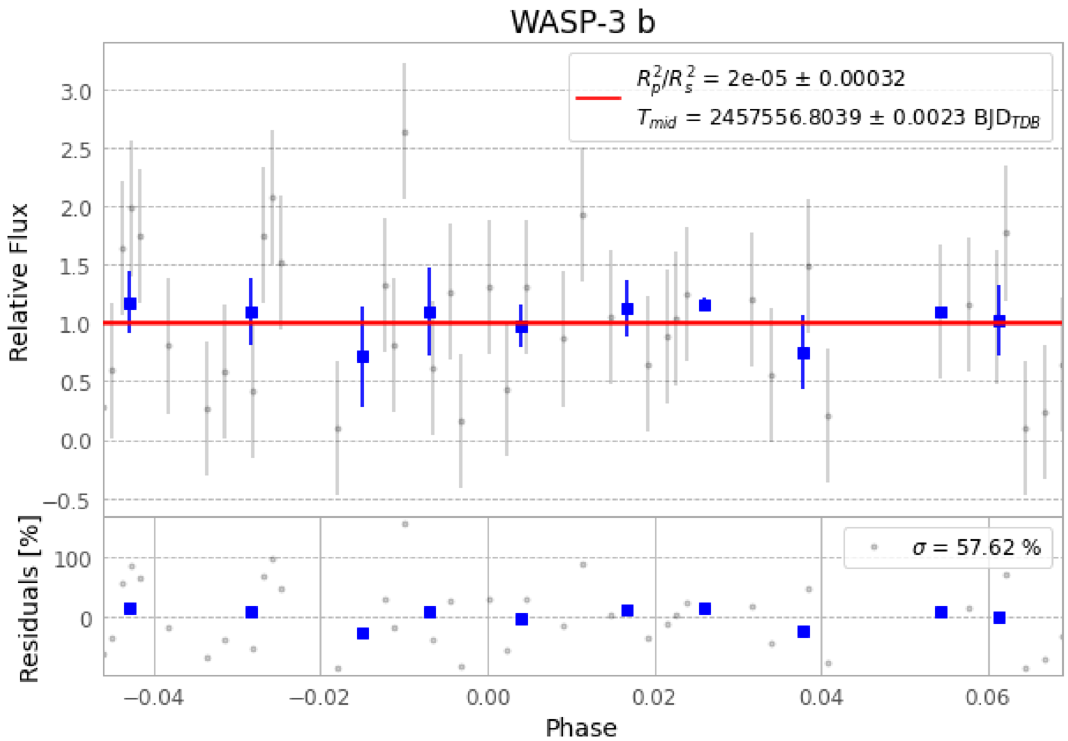

Figure 1 — FinalLightCurve_WASP-3 b_2016-06-16.png (SHA-256 5c55eb7feedc60f8…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [508, 229], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 a3f4ea476a65d139…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

110 FITS frames (72 MB), `WASP-3160617045712.FITS` → `WASP-3160617102412.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-3-160617.log (15d2585d08c4…), FinalLightCurve_WASP-3 b_2016-06-16.png (5c55eb7feedc…), FinalLightCurve_WASP-3 b_2016-06-16.csv (7c95e42fdcde…)

Compute resources

Wall time 160.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Generated 2026-07-31 10:21Z.